

similar to check valves used in plumbing systems. These allow current to flow in only one direction; the direction shown by the arrow in the diode symbol. For example, whenever A-phase voltage (voltage is similar to pressure in plumbing systems) is more positive than B- or C-phase voltages, that diode opens and allows current to flow.

When B phase becomes more positive than A phase, B-phase diode opens and A-phase diode closes. The same is true for the three diodes on the negative side of the bus. Thus, we get six current pulses as each diode opens and closes. This is called a 6-pulse VFD, which is the standard configuration for current VFDs.

We can get rid of AC ripple on DC bus by adding a capacitor. A capacitor operates in a similar fashion to a reservoir or accumulator in a plumbing system. It absorbs AC ripple and delivers smooth DC voltage.

The diode bridge converter that converts AC to DC is sometimes just referred to as a converter. The converter that converts DC back to AC is also a converter, but to distinguish it from the diode converter, it is usually referred to as an inverter. It has become common in the industry to refer to any DC-to-AC converter as an inverter.

When we close one of the top switches in the inverter, that phase of the motor is connected to the positive DC bus and voltage on that phase becomes positive. When we close one of the bottom switches in the converter, that phase is connected to the negative DC bus and becomes negative. Thus, we can make any phase on the motor positive or negative at will and can thus generate any frequency that we want. So we can make any phase positive, negative or zero.

Notice that, output from the VFD is a rectangular waveform. VFDs do not produce a sinusoidal output. This rectangular waveform would not be a good choice for a general-purpose distribution system, but is perfectly

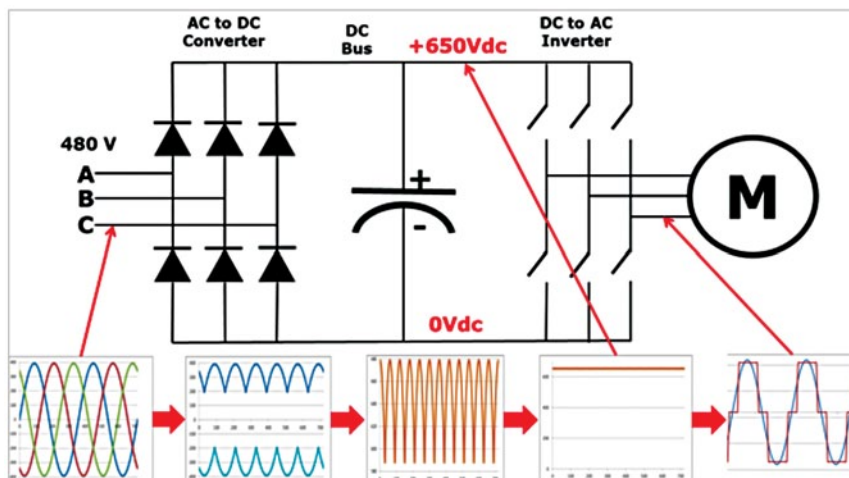


Fig. 3: Circuit model of a VFD

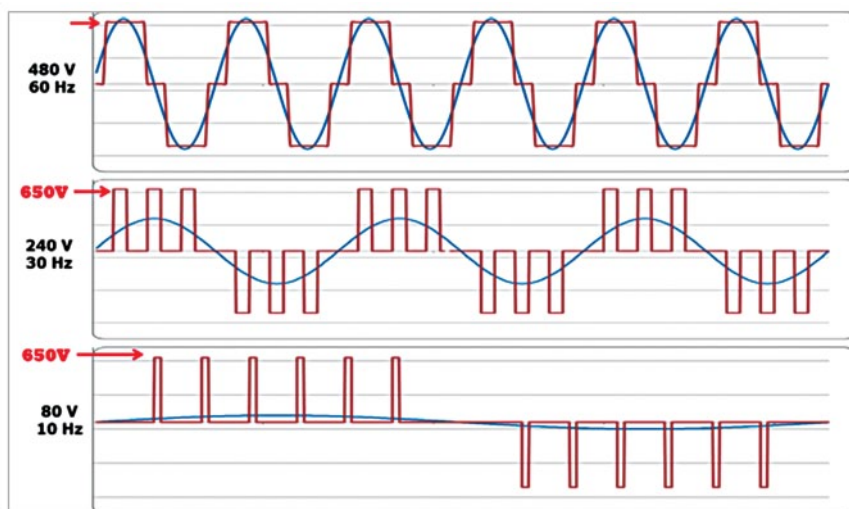


Fig. 4: Waveforms at different operating frequencies and average voltages

adequate for a motor.

If we want to reduce motor frequency, we simply switch the inverter output transistors more slowly. But if we reduce frequency, we must also reduce voltage in order to maintain V/Hz ratio. Pulse width modulation (PWM) does this.

Imagine, we could control the pressure in a water line by turning the valve on and off at high speed. While this would not be practical for plumbing systems, it works very well for VFDs.

Notice that, during the first half-cycle, voltage is on half the time and off the rest. Thus, the average voltage is half of 480V, that is, 240V. By pulsing the output, we can achieve

The specifications

- Output frequency 0.1Hz ~ 400Hz
- Adjustable V/f curve and vector control
- Carrier frequency up to 15kHz
- Auto torque boost and slip compensation
- Built-in PID feedback control
- Built-in MODBUS communication, baud rate up to 38400bps
- 0 speed holding function
- Built-in sleep/revival function
- Support communication module (DN-02, LN-01 and PD-01)

any average voltage on the output of the VFD.

Selection of Delta VFD-M as AC drive

Delta VFD-M is a sensor-less vector